

System Decomposition

From order placement to delivery confirmation — every subsystem tagged by how it splits, and every leaf a single, atomic responsibility.

10

SUBSYSTEMS

39

ATOMIC COMPONENTS

4

SPLIT TYPES

● Functional — what it does

● Process — ordered steps

● Object — which actor/entity

● Data — which information

ROOT · LEVEL 0

Food Delivery Platform System

Connects customers, restaurants, and drivers across the full order lifecycle — from browsing a menu to a confirmed hand-off at the door.



functional split → 10 subsystems

1

Customer Ordering ● process

Discovery, cart, checkout, and tracking happen in a fixed order within one order, so the split follows the process.

1.1

Discovery & Search

Browse restaurants and menus; filter by cuisine, price, ETA.

1.2

Cart Management

Add, remove, and modify items before checkout.

1.3

Checkout & Order Placement

Capture address, tip, and instructions; submit the order.

1.4

Order Tracking (Customer View)

Live map/ETA view, subscribed to the order's status feed.

2

Restaurant Catalog Management ● data

Menu, pricing, and stock are three categories of catalog data edited on their own schedule, so the split is by data.

2.1

Menu & Item Listings

Create and organize items and categories.

2.2

Item Pricing & Modifiers

Set prices and optional add-ons per item.

2.3

Stock & Inventory Flags

Mark items in or out of stock in real time.

Restaurant Order Handling ● process

Being open, accepting, and preparing form a fixed per-order sequence, so the split is by process.

3.1

Restaurant Availability Control

Toggle open/closed; throttle intake in busy-mode.

3.2

Order Acceptance / Rejection

Confirm or decline within a service window.

3.3

Kitchen Preparation Tracking

Mark prep stages; signal "ready for pickup."

Driver Dispatch & Delivery Execution ● process

Availability, matching, navigation, pickup, and confirmation happen in a fixed order for one delivery, so the split is by process.

4.1

Availability & Location Tracking

Driver on/offline state and live GPS position.

4.2

Order-Driver Matching

Assign the nearest eligible driver to a ready order.

4.3

Route Navigation & ETA

Turn-by-turn guidance and arrival estimates.

4.4

Pickup Confirmation

Driver confirms receipt at the restaurant.

4.5

Delivery Confirmation

Photo, PIN, or signature at the customer's door.

Payments & Financials ● functional

Calculating, charging, paying out, refunding, and configuring rates are distinct operations with no fixed order, so the split is functional.

5.1

Fare & Price Calculation

Item total, delivery fee, tax, and tip.

5.2

Payment Processing

Charge and authorize the customer's payment method.

5.3

Driver & Restaurant Payouts

Settle earnings on schedule.

5.4

Refunds & Disputes

Reverse or adjust charges on cancellations/complaints.

5.5

Pricing & Promotions Config

Commission, delivery-fee, and discount rules.

Notifications & Communication ● object

Each child is scoped to one recipient — customer, restaurant, or driver — so the split follows the object being messaged.

6.1

Customer Notifications

Status, ETA changes, and promo alerts.

6.2

Restaurant Notifications

New order and cancellation alerts.

6.3

Driver Notifications

New assignment and route-update alerts.

6.4

Chat & Support Messaging

Direct messaging between parties and support.

Account & Identity Management

● data

Each child owns one category of stored data — credentials, profile, roles, sessions — so the split is by data.

7.1

Registration & Authentication

Sign-up, login, credential verification.

7.2

Profile & Address Management

Personal details, saved addresses, payment methods.

7.3

Role & Permission Management

Customer, restaurant, and driver access levels.

7.4

Session & Device Management

Active sessions, tokens, registered devices.

Ratings, Feedback & Trust

● object

Each child is scoped to who or what is being rated, so the split follows the object under review.

8.1

Restaurant Ratings & Reviews

Customer feedback on food and restaurant.

8.2

Driver Ratings

Customer feedback on the delivery experience.

8.3

Customer Ratings

Restaurant/driver feedback on customer conduct.

8.4

Complaint & Issue Reporting

Intake for disputes ratings don't cover.

Order Lifecycle & State ● object

Each child manages one facet of the same object — the order itself — so the split is object-based.

9.1

Order State Machine

Valid statuses and legal transitions between them.

9.2

Order History & Audit Log

Full timeline for receipts, support, disputes.

9.3

Cancellation Handling

Who can cancel when, and the resulting side effects.

9.4

Status Change Propagation

Publishes transitions to tracking, notifications, apps.

Platform Operations, Risk & Reporting ● functional

Fraud detection, ticketing, and reporting are distinct capabilities, though grouped mainly by being internal-facing.

10.1

Fraud & Abuse Detection

Flag suspicious orders, accounts, or payments.

10.2

Support Ticketing & Escalation

Internal handling of issues raised by 8.4.

10.3

Business Intelligence & Reporting

Metrics for growth, ops, and finance.

Open questions for the room

Branches 2 & 3

Should Catalog Management and Order Handling be two subsystems, or one branch with two leaves?

Branch 9

Does Order Lifecycle & State deserve root-level peer status, or is it shared infrastructure the other branches should just reference?

Branch 10

Fraud detection, ticketing, and reporting share no real function, actor, or data — genuine subsystem, or leftover catch-all?

Every leaf above is atomic — one clear responsibility, no further split needed to stand as one implementable service or module.